

## Network-1 Project Requirements

| Project Number ( 7 )      Discussion Time ( 11:45 ) |          |                         | Team Grades (/10)  |
|-----------------------------------------------------|----------|-------------------------|--------------------|
| Team Members                                        |          |                         |                    |
| Team Member                                         | ID       | Individual grades (/10) | Total grades (/20) |
| يوسف السيد ابوزيد عبد العزيز                        | 20241150 |                         |                    |
| عمر محمد جلال عبد الوهاب                            | 20240640 |                         |                    |
| مريم عبد الباقي محمد ماهر                           | 20240939 |                         |                    |
| مريم صابر لبيب سلوانس                               | 20240936 |                         |                    |
| مريم محمد عبد المحسن عبد الغفار                     | 20240948 |                         |                    |
| هند مصطفى سعد مصطفى                                 | 20241105 |                         |                    |
| منة محمود احمد                                      | 20241022 |                         |                    |
|                                                     |          |                         |                    |

### Network-1 Project Requirements

| Requirement             | Details                                                       | Marks | Grade |
|-------------------------|---------------------------------------------------------------|-------|-------|
| Network Design          | All networks reachable and used right devices                 | 1     |       |
| VLSM                    | Correct VLSM calculations                                     | 1     |       |
|                         | Complete addressing table (Network, First/Last IP, Broadcast) | 1     |       |
| Protocol Implementation | DHCP                                                          | 1     |       |
|                         | DNS Server                                                    | 1     |       |
|                         | VLAN                                                          | 1     |       |
|                         | NAT (Static/PAT)                                              | 1     |       |
|                         | VPN                                                           | 1     |       |
|                         | Static Routing                                                | 1     |       |
|                         | Web server                                                    | 1     |       |

### Network-1 Project Documentation

| Requirement    | What You Must Do                                                                           | How to Verify (Commands) | Required Screenshots                             |
|----------------|--------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------|
| Network Design | Design a WAN using at least 4 routers, including PCs, switches, DNS server, and Web server | ping                     | Full topology + successful ping between networks |

|                         |                                                                                                                  |                                                    |                                             |
|-------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------|
| <b>VLSM Calculation</b> | Perform subnetting based on number of hosts needed                                                               | —                                                  | VLSM calculation table                      |
| <b>Addressing Table</b> | Provide Network ID, Subnet Mask, First IP, Last IP, Broadcast for each subnet                                    | show ip interface brief                            | Router IP configurations                    |
| <b>DHCP</b>             | Configure router to assign IP addresses automatically to PCs                                                     | show ip dhcp pool                                  | DHCP + sample PC IP settings                |
| <b>DNS Server</b>       | Configure domain name (e.g., <a href="http://www.company.com">www.company.com</a> ) and link it to Web Server IP | —                                                  | Browser accessing website using domain      |
| <b>VLAN</b>             | Create VLANs and assign switch ports to each VLAN                                                                | show vlan brief                                    | VLAN table                                  |
| <b>NAT (Static/PAT)</b> | Configure Static NAT for servers and PAT or Dynamic NAT for users                                                | show ip nat translations<br>show ip nat statistics | NAT translation table                       |
| <b>VPN</b>              | Configure VPN (Site-to-Site) between networks                                                                    | show crypto isakmp sa<br>show crypto ipsec sa      | VPN status + successful ping                |
| <b>Static Routing</b>   | Configure static routes between all routers                                                                      | show ip route                                      | Routing table from each router              |
| <b>Web Server</b>       | Create and host a simple HTML page                                                                               | —                                                  | Web page opened in browser                  |
| <b>ICMP Testing</b>     | Test connectivity between all networks                                                                           | ping                                               | Successful ping + simulation mode scenarios |

### Final Submission Checklist

| Item                      | Requirement                                          |
|---------------------------|------------------------------------------------------|
| <b>Packet Tracer File</b> | Submit the .pkt file                                 |
| <b>Report</b>             | Include all configurations, tables, and explanations |
| <b>Screenshots</b>        | Show commands and their outputs clearly              |
| <b>Functionality</b>      | Ensure all networks and services are working         |

### Important Project Day Instructions

- Be on time; late arrival will result in loss of marks.
- Bring your printed documentation report and cover sheet; missing documents will result in loss of marks.
- Bring your laptop ready with the Packet Tracer file (.pkt); without it, you cannot present your project.
- All team members must understand every part of the project; anyone can be asked any question on project implementation and course concepts.
- Be polite and respectful during the discussion and the moderator has full authority to deduct marks for any inappropriate behavior.